

Exp No:6 Date:	Gradient Descent Method- Stochastic Gradient Descent Algorithm
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AIM:

Stochastic Gradient Descent (SGD) iteratively updates model parameters using individual training examples to minimize prediction error, enabling faster convergence and efficient handling of large datasets.

PROCEDURE:

Initialization: Set weights to zero. Use feature matrix X and target vector y . Define learning rate and number of epochs.

Gradient Computation: Compute gradient for each sample as $(\text{prediction} - \text{target}) \times \text{feature}$.

Weights Update: Update weights using the negative gradient scaled by the learning rate.

Problem-1:

X	Y
1	2
2	3
3	6
4	8

Code:

```
import numpy as np

def gradient_descent(x, y, lr=0.01, epochs=100): m, c = 0, 0
    n = len(x)

    for _ in range(epochs):
        y_pred = m * x + c

        dm = (-2/n) * sum(x * (y - y_pred)) dc = (-2/n) * sum(y - y_pred)

        m = m - lr * dm
        c = c - lr * dc

    print("Slope:", m)
    print("Intercept:", c)

    x = np.array([1, 2, 3, 4])
    y = np.array([2, 4, 6, 8])

    gradient_descent(x, y)
```

Output:

```
Slope: 2.0
Intercept: 0.0
```

RESULT:

Thus, the Gradient Descent Method was successfully executed and verified using Python.